TDI & Representations

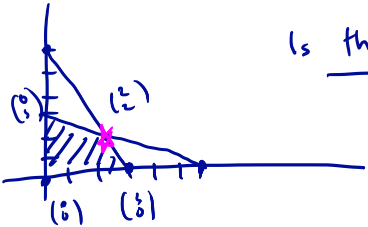
(A, b) TDI, $b \in \mathbb{Z}^m$

$\Rightarrow \{x : Ax \le b\}$ is integral.

Converse need not hold.

An Example — 1/2

$$P = \{ x_1, x_2 \geq 0, \ x_1 + 2x_2 \leq 6, \ 2x_1 + x_2 \leq 6 \}$$



Is this TDI ?

$$\max: \ c^T x = x_1 + x_2$$

$$\text{s.t.: } \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} x \leq \begin{pmatrix} 6 \\ 6 \\ 0 \\ 0 \end{pmatrix}$$

$$\min: \ 6y_1 + 6y_2$$

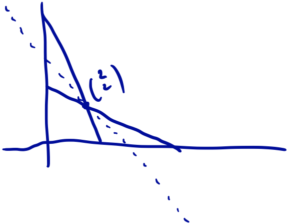
$$\text{s.t. } \begin{aligned} y_1 + 2y_2 - y_3 &= 1 \\ 2y_1 + y_2 - y_4 &= 1 \\ y_i &\geq 0 \end{aligned}$$

$x^* = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$, $y^* = \begin{pmatrix} 1/3 \\ 1/3 \\ 0 \\ 0 \end{pmatrix}$ opt solution

Not TDI.

An Example — 2/2

$$P = \{ x_1, x_2 \geq 0,\ 2x_1 + x_2 \leq 6,\ x_1 + 2x_2 \leq 6,\ x_1 + x_2 \leq 4 \}$$



$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 6 \\ 6 \\ 0 \\ 0 \\ 4 \end{pmatrix}$$

Dual: min: $6y_1 + 6y_2 + 4y_5$

st: $y_1 + 2y_2 - y_3 + y_5 = 1$

$2y_1 + y_2 - y_4 + y_5 = 1$

$y_i \geq 0$

$$y^* = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

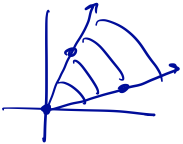
Hilbert Basis

A H.B. is a set $\{a_1, \dots, a_N\}$

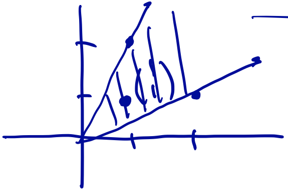
s.t. $\forall b \in \text{cone}\{a_1, \dots, a_N\}$ and

b is integral, b is a nonneg integral comb. of $a_1, \dots, a_N$.

[Recall: $C = \text{cone}\{v_1, \dots, v_k\} = \{x = \sum \lambda_i v_i \, , \; \lambda_i \geq 0\}$]



Ex: $C = \text{Cone}\left\{\begin{pmatrix}2\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}\right\} =$



Not a HB.

$C = \text{cone}\left\{\begin{pmatrix}2\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\}$

Is a HB.